



Project Title:



Upgrading Amazon EBS Volume from io1 to io2 Without Downtime



Problem Statement

A software company wants to **upgrade its EBS Provisioned IOPS SSD volumes (**)** to the newer **** volumes** to benefit from higher durability and lower cost — **without interrupting running EC2 workloads or risking data loss.**



Objective

- Launch an EC2 instance with an io1 volume.
 - Connect to the instance via SSH.
 - Upgrade the EBS volume type from io1 to io2 using **in-place modification** (no downtime).
 - Verify and optionally benchmark volume performance.
 - Safely clean up resources to avoid unnecessary charges.
-



Tools & Technologies

Tool/Service	Purpose
Amazon EC2	Virtual server for testing
Amazon EBS	Block storage volumes (io1, io2)
AWS Management Console	GUI to manage EC2 and EBS
PowerShell (Windows)	SSH into EC2 instance
Ubuntu (EC2 AMI)	Linux OS for testing environment
fio (Linux tool)	Optional performance testing



Step-by-Step Implementation



1. Launch EC2 Instance with io1 Volume

- Chose **Ubuntu AMI** and t2.micro instance type
- Customized EBS root volume:
 - Type: Provisioned IOPS SSD (io1)

- Size: 10 GiB
 - IOPS: 100
 - Downloaded .pem key pair: ebs-volume-transfer.pem
-

☒ 2. Connect to EC2 via SSH (PowerShell on Windows)

```
cd "C:\Users\<YourUser>\Downloads"
```

```
ssh -i .\ebs-volume-transfer.pem ubuntu@<your-ec2-public-ip>
```

☒ 3. Verify Disk Inside EC2

```
lsblk
```

Confirmed the xvda disk is the root volume.

☒ 4. Modify Volume from io1 → io2 (No Downtime)

- Go to **EC2 → Volumes**
- Select the attached volume
- Click **Actions → Modify volume**
- Changed:
 - **Volume type:** io2
 - Kept size and IOPS the same
- Clicked **Modify → Yes**

☒ AWS modified the volume **in-place while the instance remained online.**

☒ 5. (Optional) Run Disk Benchmark with fio

```
sudo apt update && sudo apt install -y fio
```

```
fio --name=test --ioengine=libaio --rw=randread --bs=4k --direct=1 --size=1G  
--numjobs=1 --runtime=60 --group_reporting
```

☒ 6. Clean Up

- Terminated EC2 instance
 - Deleted EBS volume and .pem key
 - (Optional) Created and deleted snapshot
-

✓ Outcome

- Successfully transitioned from io1 to io2 EBS volume **without stopping the EC2 instance or losing data**.
- Verified volume remained mounted and functional throughout.

🧠 Key Learnings

- io2 volumes provide better durability (99.999%) and cost efficiency.
- AWS allows in-place volume modification without downtime.
- Hands-on EC2 and EBS experience strengthens understanding of real-world cloud migration scenarios.

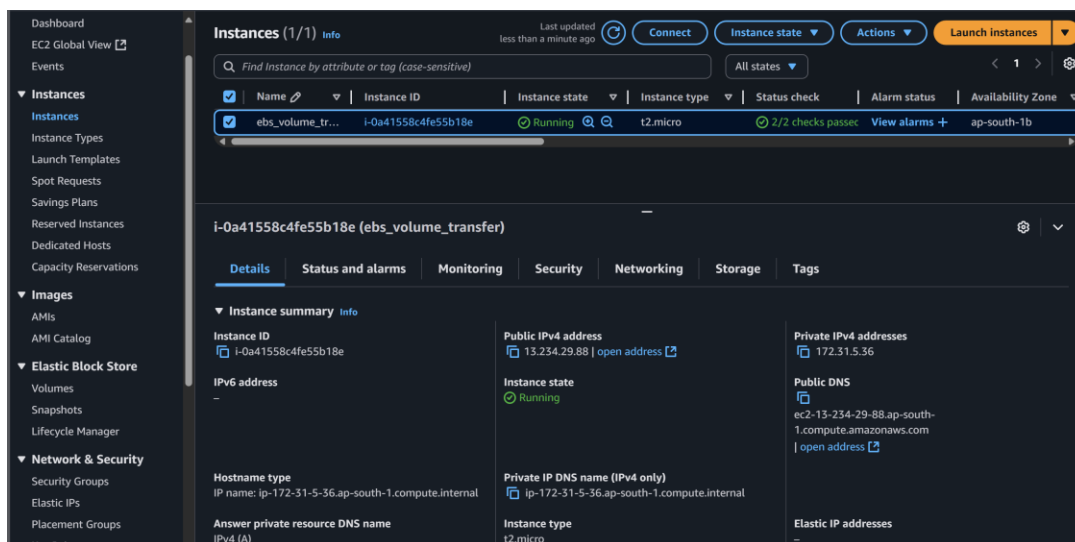
🔗 Useful Commands Summary

Task	Command
SSH into EC2	<code>ssh -i key.pem ubuntu@ip</code>
Check disk	<code>lsblk</code>
Install fio	<code>sudo apt install -y fio</code>
Run IOPS test	<code>fio --name=test ...</code>

📁 Workspace Output (Paste Your Results Below)

📄 Paste screenshots, terminal logs, AWS console screenshots, and fio benchmark outputs below:

[EC2 instance screen after SSH login]



The screenshot displays the AWS Management Console interface for an EC2 instance. The left sidebar shows navigation options like Dashboard, EC2 Global View, Events, and Instances. The main panel shows the 'Instances (1/1)' page with a table listing the instance 'ebs_volume_transfer' (ID: i-0a41558c4fe55b18e) in a 'Running' state. Below the table, the 'Details' tab is selected, showing the instance summary with fields like Instance ID, Public IPv4 address (13.234.29.88), Private IPv4 addresses (172.31.5.36), Public DNS (ec2-13-234-29-88.ap-south-1.compute.amazonaws.com), Private DNS name (ip-172-31-5-36.ap-south-1.compute.internal), and Instance type (t2.micro).

[lsblk output screenshot]

```
ubuntu@ip-172-31-5-36:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0        7:0    0 27.2M  1 loop /snap/amazon-ssm-agent/11320
loop1        7:1    0 73.9M  1 loop /snap/core22/1981
loop2        7:2    0 50.9M  1 loop /snap/snapd/24505
xvda         202:0    0   8G  0 disk
├─xvda1      202:1    0    7G  0 part /
├─xvda14     202:14   0    4M  0 part
├─xvda15     202:15   0 106M  0 part /boot/efi
└─xvda16     259:0    0 913M  0 part /boot
```

[Volume modification success screenshot]

Modify volume [Info](#)
Modify the type, size, and performance of an EBS volume.

Volume details

Volume ID
[vol-0af0d6942847d7307](#)

Volume type [Info](#)
Provisioned IOPS SSD (io2)

Size (GiB) [Info](#)
10
Min: 4 GiB, Max: 65536 GiB.

IOPS [Info](#)
3000
Min: 100 IOPS, Max: 10000 IOPS (up to 1000 IOPS per GiB)

Amazon EBS Multi-Attach
☐ Enable Multi-Attach

[Cancel](#) [Modify](#)

[fio performance test results]

```
ubuntu@ip-172-31-5-36:~$ sudo apt update && sudo apt install -y fio
fio --name=test --ioengine=libaio --rw=randread --bs=4k --direct=1 --size=1G --numjobs=1 --runtime=60 --group_reporting
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
23 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
fio is already the newest version (3.36-1ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 23 not upgraded.
test: (g=0): rw=randread, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=libaio, iodepth=1
fio-3.36
Starting 1 process
test: Laying out IO file (1 file / 1024MiB)
Jobs: 1 (f=1): [r(1)][100.0%][r=9657KiB/s][r=2414 IOPS][eta 00m:00s]
test: (groupid=0, jobs=1): err= 0: pid=2234: Mon Jun 23 23:31:18 2025
read: IOPS=2337, BW=9351KiB/s (9575kB/s)(548MiB/60001msec)
  slat (usec): min=10, max=1354, avg=15.96, stdev=14.29
  clat (nsec): min=1546, max=4838.1k, avg=409055.31, stdev=206611.03
  lat (usec): min=123, max=4850, avg=425.02, stdev=207.45
  clat percentiles (usec):
    | 1.00th=[ 129], 5.00th=[ 149], 10.00th=[ 200], 20.00th=[ 265],
    | 30.00th=[ 306], 40.00th=[ 310], 50.00th=[ 318], 60.00th=[ 338],
    | 70.00th=[ 502], 80.00th=[ 578], 90.00th=[ 701], 95.00th=[ 816],
    | 99.00th=[ 1037], 99.50th=[ 1123], 99.90th=[ 1303], 99.95th=[ 1434],
    | 99.99th=[ 1745]
  bw ( KiB/s): min= 8304, max=19016, per=100.00%, avg=9353.88, stdev=1000.52, samples=119
  iops : min= 2076, max= 4754, avg=2338.47, stdev=250.13, samples=119
  lat (usec) : 2=0.01%, 4=0.01%, 10=0.01%, 20=0.01%, 50=0.01%
  lat (usec) : 100=0.01%, 250=15.75%, 500=54.10%, 750=22.59%, 1000=6.17%
  lat (msec) : 2=1.35%, 4=0.01%, 10=0.01%
  cpu      : usr=1.49%, sys=4.97%, ctx=140241, majf=0, minf=15
IO depths : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwts: total=10262,0,0,0 short=0,0,0 dropped=0,0,0
latency   : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=9351KiB/s (9575kB/s), 9351KiB/s-9351KiB/s (9575kB/s-9575kB/s), io=548MiB (575MB), run=60001-60001msec
```

Tags

#AWS #EBSUpgrade #io1toio2 #InPlaceModification #EC2 #RealWorldMigration
#NoDowntime #CloudProject

Author & Date

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